REPORT

Reviewer report: pytest-dev/pytest#15380: Fix

INTERNALERROR in --fixtures when two fixture defs

share a sort key

Run pytest-15380, tested 2026-11-12 11:24 IST by ops/daily_run.py (route source-pair). Proof-of-Change

commit ef12096677e9cf59e8673436d74ec65d53771d7c. Report generated 2026-11-12 11:56 IST by

reports/pr_report.py.

Baseline: full (test comparison + claim probes + coverage gate + claim flagging (differential probes shown,

informational)).

COULD NOT VERIFY

Recommended action: Re-run Confidence: Low

The run did not produce a usable comparison, so there is no conclusion about the PR. The run did not finish

cleanly, so it says nothing about the PR.

Advice only: Proof-of-Change never blocks a merge and never posts on GitHub.

Summary in plain English

• What it says it does: Fix INTERNALERROR in --fixtures when two fixture defs share a sort key

(pytest-dev/pytest#15380).

• What we observed: tests produced no results on both sides (collection failed; see pytest logs); probe

model-probe1 is not_adequate (ran none of the 1 changed line(s) on the main side); its results do not count

either way.

• What you should do: Do not decide yet. Fix the problem under 'How to fix' and run the check again.

Demo helper

These notes are background from the console and the daily check. They do not change the verdict.

If setup fails, the helper retries the install up to 2 times. If a test still can't be verified because something is

missing or fails to import, it looks up the cause, installs the fix on both sides and reruns that step, up to 3 tries.

• Setup check: GitHub token: present.

• Setup check: Nebius token: present.

• Setup check: free memory 8.5 GB (of 15.6 GB).

• Setup check: free disk 91.8 GB.

• Demo helper: git is installed.

• Demo helper: python3 is installed.

• Demo helper: uv is installed.

• Looked up with Tavily: [redacted] results

• Query: pytest python setup install instructions https://github.com/pytest-dev/pytest

site:github.com/pytest-dev/pytest

• Query: pytest python documentation install https://github.com/pytest-dev/pytest

site:github.com/pytest-dev/pytest

• Query: pytest python install requirements https://pypi.org/project/pytest/
https://github.com/pytest-dev/pytest

site:pypi.org

• Source: pytest-testinfra · PyPI — https://pypi.org/project/pytest-testinfra/

• Source: pytest/CONTRIBUTING.rst at main · pytest-dev/pytest · GitHub —

https://github.com/pytest-dev/pytest/blob/main/CONTRIBUTING.rst

• Source: GitHub - pytest-dev/pytest-runner · GitHub — https://github.com/pytest-dev/pytest-runner

• Source: pytest has no documented way for the user to point it at local source files. · Issue #1937 ·

pytest-dev/pytest · GitHub — https://github.com/pytest-dev/pytest/issues/1937

• Source: GitHub - pytest-dev/pytest: The pytest framework makes it easy to write small tests, yet scales to

support complex functional testing · GitHub — https://github.com/pytest-dev/pytest

• Source: Releases · pytest-dev/pytest · GitHub — https://github.com/pytest-dev/pytest/releases

• Source: PyPI recent updates for pytest — https://pypi.org/rss/project/pytest/releases.xml

• Setup plan: the same packages on main and pr.

• Setup plan: colorama is noted, installed with the project.

• Setup plan: exceptiongroup is noted, installed with the project.

• Setup plan: iniconfig is noted, installed with the project.

• Setup plan: packaging is noted, installed with the project.

• Setup plan: pluggy is noted, installed with the project.

• Setup plan: pygments is noted, installed with the project.

• Setup plan: tomli is noted, installed with the project.

• Setup plan: install argcomplete (listed in the dev extra) on main and pr.

• Setup plan: install asynctest; python_version<'3.11' (listed in the dev extra) on main and pr.

• Setup plan: install attrs>=23.1 (listed in the dev extra) on main and pr.

• Setup plan: install coverage>=7.5 (listed in the dev extra) on main and pr.

• Setup plan: install hypothesis>=6.75 (listed in the dev extra) on main and pr.

• Setup plan: install mock (listed in the dev extra) on main and pr.

• Setup plan: install numpy>=1.26 (listed in the dev extra) on main and pr.

• Setup plan: install pexpect>=4.9 (listed in the dev extra) on main and pr.
• Setup plan: install pytest-xdist>=3.5 (listed in the dev extra) on main and pr.
• Setup plan: install pyyaml (listed in the dev extra) on main and pr.
• Setup plan: install requests (listed in the dev extra) on main and pr.
• Setup plan: install setuptools (listed in the dev extra) on main and pr.
• Setup plan: install xmlschema (listed in the dev extra) on main and pr.
• Setup plan: install pygments-pytest>=2.5.0 (listed in requirements.txt) on main and pr.
• Setup plan: install sphinx-removed-in>=0.2.0 (listed in requirements.txt) on main and pr.
• Setup plan: install sphinx>=9 (listed in requirements.txt) on main and pr.
• Setup plan: install sphinxcontrib-trio>=1.2.0 (listed in requirements.txt) on main and pr.
• Setup plan: install sphinxcontrib-svg2pdfconverter (listed in requirements.txt) on main and pr.
• Setup plan: install furo (listed in requirements.txt) on main and pr.
• Setup plan: install sphinxcontrib-towncrier (listed in requirements.txt) on main and pr.
• Setup plan: install sphinx-issues (listed in requirements.txt) on main and pr.
• Setup plan: install sphinx-inline-tabs (listed in requirements.txt) on main and pr.

• Setup plan: install anyio[trio]==4.15.1 (listed in requirements.txt) on main and pr.
• Setup plan: install django==6.1.2 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-asyncio==1.4.1.post1 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-bdd==9.0.0 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-cov==7.1.0 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-django==4.14.0 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-flakes==4.0.5 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-html==4.2.0 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-mock==3.17.0 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-rerunfailures==16.7 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-sugar==1.1.1 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-trio==0.8.0 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-twisted==1.14.3 (listed in requirements.txt) on main and pr.
• Setup plan: install twisted==26.10.0 (listed in requirements.txt) on main and pr.
• Setup plan: install pytest-xvfb==3.1.1 (listed in requirements.txt) on main and pr.
• Setup plan: install pre-commit>=4 (listed in tox.ini deps) on main and pr.
• Setup plan: install regendoc>=0.8.1 (listed in tox.ini deps) on main and pr.
• Setup plan: install pyinstaller (listed in tox.ini deps) on main and pr.
• Setup plan: install towncrier (listed in tox.ini deps) on main and pr.
• Setup plan: install pypandoc_binary (listed in tox.ini deps) on main and pr.
• Setup plan: install tabulate[widechars] (listed in tox.ini deps) on main and pr.

• Setup plan: install tqdm (listed in tox.ini deps) on main and pr.

• Setup plan: install requests-cache (listed in tox.ini deps) on main and pr.

• Setup plan: install platformdirs (listed in tox.ini deps) on main and pr.

• Setup plan: Python 3.12: Python 3.12 already meets >=3.10

• Installed argcomplete 3.7.2 from pyproject (listed in the dev extra) on main and pr.

• Installed asynctest from pyproject (listed in the dev extra) on main and pr.

• Installed attrs 26.1.0 from pyproject (listed in the dev extra) on main and pr.

• Installed coverage 7.17.2 from pyproject (listed in the dev extra) on main and pr.

• Installed hypothesis 6.170.15 from pyproject (listed in the dev extra) on main and pr.

• Installed mock 5.2.0 from pyproject (listed in the dev extra) on main and pr.

• Installed numpy 2.5.4 from pyproject (listed in the dev extra) on main and pr.

• Installed pexpect 4.9.0 from pyproject (listed in the dev extra) on main and pr.

• Installed pytest-xdist 3.8.0 from pyproject (listed in the dev extra) on main and pr.

• Installed pyyaml 6.0.3 from pyproject (listed in the dev extra) on main and pr.

• Installed requests 2.34.2 from pyproject (listed in the dev extra) on main and pr.

• Installed setuptools 84.2.1 from pyproject (listed in the dev extra) on main and pr.

• Installed xmlschema 4.3.2 from pyproject (listed in the dev extra) on main and pr.

• Installed pygments-pytest 2.5.0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed sphinx-removed-in 0.2.3 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed sphinx 9.1.1 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed sphinxcontrib-trio 1.2.0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed sphinxcontrib-svg2pdfconverter 2.1.0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed furo 2025.12.19 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed sphinxcontrib-towncrier 0.5.0a0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed sphinx-issues 6.0.0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed sphinx-inline-tabs 2025.12.21.14 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed anyio 4.15.1 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed django 6.1.2 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-asyncio 1.4.1.post1 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-bdd 9.0.0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-cov 7.1.0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-django 4.14.0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-flakes 4.0.5 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-html 4.2.0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-mock 3.17.0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-rerunfailures 16.7 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-sugar 1.1.1 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-trio 0.8.0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-twisted 1.14.3 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed twisted 26.10.0 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pytest-xvfb 3.1.1 from requirements.txt (listed in requirements.txt) on main and pr.

• Installed pre-commit 4.7.0 from tox.ini (listed in tox.ini deps) on main and pr.

• Installed regendoc 0.9.0 from tox.ini (listed in tox.ini deps) on main and pr.

• Installed pyinstaller 6.22.3 from tox.ini (listed in tox.ini deps) on main and pr.

• Installed towncrier 26.9.0 from tox.ini (listed in tox.ini deps) on main and pr.

• Installed pypandoc_binary 1.17 from tox.ini (listed in tox.ini deps) on main and pr.

• Installed tabulate 0.10.0 from tox.ini (listed in tox.ini deps) on main and pr.

• Installed tqdm 4.70.1 from tox.ini (listed in tox.ini deps) on main and pr.

• Installed requests-cache 1.3.3 from tox.ini (listed in tox.ini deps) on main and pr.

• Installed platformdirs 4.14.5 from tox.ini (listed in tox.ini deps) on main and pr.

Setup: detection fired. Pytest stopped at startup the same way on both sides. Plugin chosen: asyncio. The

installed pytest plugin list was read. Retry flags: -p no:asyncio. Outcome: failed again. Main:

pytest.PytestDeprecationWarning: The configuration option "asyncio_default_fixture_loop_scope" is unset.. PR:

pytest.PytestDeprecationWarning: The configuration option "asyncio_default_fixture_loop_scope" is unset.. The

first result was kept. This does not change how the verdict is chosen.

Claims vs reality

| # | Claim (source) | Status | What we observed | Log line

| P-C1 | Running pytest --fixtures crashes with an INTERNALERROR when two fixture definitions share every

display sort key. (PR description) | not observed | probe model-probe1, c1.id_success: same on both sides: false

| probe-model-probe1-pr-1.jsonl:2 «"observation": "c1.id_success"»

| P-C2 | The fix sorts fixture definition entries by display keys only using key=lambda entry: entry[:-1]. (PR

description) | not observed | probe model-probe1, c2.key_exception: same on both sides: null |

probe-model-probe1-pr-1.jsonl:3 «"observation": "c2.key_exception"»

| P-C3 | Dropping the trailing FixtureDef from the sort comparison is safe because it is used only for rendering,

not for ordering, and removes the only non■comparable element. (PR description) | proven | probe

model-probe1, c3.fixturedef_used_for_rendering: same on both sides: false | probe-model-probe1-pr-1.jsonl:6

«"observation": "c3.fixturedef_used_for_rendering"»

| P-C4 | The regression test registers two FixtureDefs wrapping the same function under equal■length baseids,

runs --fixtures, and shows that it dies with INTERNALERROR on the current source but lists both fixtures

normally after the fix. (PR description) | not observed | probe model-probe1, c4.sort_fails_without_key: same on

both sides: true | probe-model-probe1-pr-1.jsonl:7 «"observation": "c4.sort_fails_without_key"»

| P-C5 | After applying the fix, the regression test in test_helpconfig.py passes (test 16). (PR description) | not

observed | probe model-probe1, c5.test_passes: same on both sides: null | probe-model-probe1-pr-1.jsonl:9

«"observation": "c5.test_passes"»

| P-C6 | The repository's ruff lint and format checks pass after the fix. (PR description) | proven | probe

model-probe1, c6.ruff_pass: same on both sides: null | probe-model-probe1-pr-1.jsonl:10 «"observation":

"c6.ruff_pass"»

Citation check (PASS): 6 citation(s), every one re-read from its log file and matched.

What changed in behavior

Not measured: tests produced no results on both sides (collection failed; see pytest logs); probe model-probe1 is

not_adequate (ran none of the 1 changed line(s) on the main side); its results do not count either way.

Probes

Model-written probes: model provider: nebius (Token Factory, live).

A probe is a small deterministic script run on both sides in the same environments as the tests, 3 alternating runs

per side. It prints JSON observations; plain code (poc/diff.py) compares them. A difference that no claim of the

PR covers needs attention. Every probe was frozen (content sha256) before any PR-side code ran, and checked

against that freeze before each PR-side run. Only a live model probe is blind; fixed probes and recorded model

fixtures are known-answer cases.

model-probe1 (model (nvidia/nemotron-3-super-120b-a12b, live), model-probes/model-probe1.py, frozen

8797a378db0b at 2026-11-12T05:58:59.029882+00:00): Probe 1: Checks the fix in _showfixtures_main and

tests sorting keys.

• P-C1 (PR description): Running pytest --fixtures crashes with an INTERNALERROR when two fixture

definitions share every display sort key. Expects the covered observations to differ.

• P-C2 (PR description): The fix sorts fixture definition entries by display keys only using key=lambda entry:

entry[:-1]. Expects the covered observations to differ.

• P-C3 (PR description): Dropping the trailing FixtureDef from the sort comparison is safe because it is used only

for rendering, not for ordering, and removes the only non■comparable element. Expects the covered

observations to stay the same.

• P-C4 (PR description): The regression test registers two FixtureDefs wrapping the same function under

equal■length baseids, runs --fixtures, and shows that it dies with INTERNALERROR on the current source but

lists both fixtures normally after the fix. Expects the covered observations to differ.

• P-C5 (PR description): After applying the fix, the regression test in test_helpconfig.py passes (test 16). Expects

the covered observations to differ.

• P-C6 (PR description): The repository's ruff lint and format checks pass after the fix. Expects the covered

observations to stay the same.

| Observation | Claim | Result | What the probe saw (main -> PR) | Log lines (main, PR)

| c1.id_exception | P-C1 | probe could not run | same on both sides: "TypeError" |

probe-model-probe1-main-1.jsonl:1 «"observation": "c1.id_exception"»
probe-model-probe1-pr-1.jsonl:1

«"observation": "c1.id_exception"»

| c1.id_success | P-C1 | no change although the claim says it changes | same on both sides: false |

probe-model-probe1-main-1.jsonl:2 «"observation": "c1.id_success"»
probe-model-probe1-pr-1.jsonl:2

«"observation": "c1.id_success"»

| c2.key_exception | P-C2 | no change although the claim says it changes | same on both sides: null |

probe-model-probe1-main-1.jsonl:3 «"observation": "c2.key_exception"»
probe-model-probe1-pr-1.jsonl:3

«"observation": "c2.key_exception"»

| c2.key_success | P-C2 | no change although the claim says it changes | same on both sides: true |

probe-model-probe1-main-1.jsonl:4 «"observation": "c2.key_success"»
probe-model-probe1-pr-1.jsonl:4

«"observation": "c2.key_success"»

| c2.uses_fix | P-C2 | as claimed | value: false -> true | probe-model-probe1-main-1.jsonl:5 «"observation":

"c2.uses_fix"»
probe-model-probe1-pr-1.jsonl:5 «"observation": "c2.uses_fix"»

| c3.fixturedef_used_for_rendering | P-C3 | as claimed | same on both sides: false |

probe-model-probe1-main-1.jsonl:6 «"observation":

"c3.fixturedef_used_for_rendering"»
probe-model-probe1-pr-1.jsonl:6 «"observation":

"c3.fixturedef_used_for_rendering"»

| c4.sort_fails_without_key | P-C4 | no change although the claim says it changes | same on both sides: true |

probe-model-probe1-main-1.jsonl:7 «"observation":

"c4.sort_fails_without_key"»
probe-model-probe1-pr-1.jsonl:7 «"observation": "c4.sort_fails_without_key"»

| c4.sort_succeeds_with_key | P-C4 | no change although the claim says it changes | same on both sides: true |

probe-model-probe1-main-1.jsonl:8 «"observation":

"c4.sort_succeeds_with_key"»
probe-model-probe1-pr-1.jsonl:8 «"observation":

"c4.sort_succeeds_with_key"»

| c5.test_passes | P-C5 | no change although the claim says it changes | same on both sides: null |

probe-model-probe1-main-1.jsonl:9 «"observation": "c5.test_passes"»
probe-model-probe1-pr-1.jsonl:9

«"observation": "c5.test_passes"»

| c6.ruff_pass | P-C6 | as claimed | same on both sides: null | probe-model-probe1-main-1.jsonl:10

«"observation": "c6.ruff_pass"»
probe-model-probe1-pr-1.jsonl:10 «"observation": "c6.ruff_pass"»

Citation check (PASS): 20 citation(s), every one re-read from its log file and matched.

Was the changed code exercised?

Coverage: NOT MEASURED (coverage has not been run for this check).

Coverage gate (python, confidence lower): coverage of the changed Python lines was NOT MEASURED

(coverage has not been run for this check).

How this is measured: one extra, untimed run of the PR side's tests under coverage.py. Changed lines come

from comparing the tested main and PR files. Compiled code (C, C++, Cython) is invisible to coverage.py; such a

change only counts as verified when the PR adds or changes a test for that module.

What went right

• Tested e219fd7f5d66 of the PR against main (source-pair route).

• Probes ran on both sides: 1 probe(s), 3 alternating runs per side, every run printed its observations.

• Single-thread environment: OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0.

• Machine type recorded: local x86_64 8 cpus.

• Nothing posted upstream: GitHub was only read.

• Known-answer replay (fresh re-run): replayed 2026-10-10 06:30:21 IST, harness 1.0.

• Every claim cites a real log line: 6 citation(s), every one re-read from its log file and matched.

• Every probe observation cites its output line: 20 citation(s), every one re-read from its log file and matched.

What went wrong and how to fix it

| Issue | Status | Detail | How to fix

| Run did not finish | COULD NOT VERIFY | tests produced no results on both sides (collection failed; see pytest

logs); probe model-probe1 is not_adequate (ran none of the 1 changed line(s) on the main side); its results do

not count either way | Read the setup/build log under Evidence links; build failures usually need a compiler or a

newer nightly.

| 3 alternating runs per side | FAIL | main 3, PR 3 of 0 | Re-run with the default --runs 3 and look for timeouts in

the pytest logs.

| Changed code was exercised (coverage gate) | WARN | coverage of the changed Python lines was NOT

MEASURED (coverage has not been run for this check) | The coverage run did not produce data (see

coverage.why); run the check again.

Evals and guardrails

| Check | Result | Detail

| 3 alternating runs per side | FAIL | main 3, PR 3 of 0

| Changed code was exercised (coverage gate) | WARN | coverage of the changed Python lines was NOT

MEASURED (coverage has not been run for this check)

| Probes ran on both sides | PASS | 1 probe(s), 3 alternating runs per side, every run printed its observations

| Single-thread environment | PASS | OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1,

MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1,

BLIS_NUM_THREADS=1, PYTHONHASHSEED=0

| Machine type recorded | PASS | local x86_64 8 cpus

| Nothing posted upstream | PASS | GitHub was only read

| Known-answer replay (fresh re-run) | PASS | replayed 2026-10-10 06:30:21 IST, harness 1.0

| Every claim cites a real log line | PASS | 6 citation(s), every one re-read from its log file and matched

| Every probe observation cites its output line | PASS | 20 citation(s), every one re-read from its log file and

matched

Checklist

- I read the verdict and the plain-English summary.
- Every 'false' or 'unmentioned change' row has a log line I can open.
- Nothing under 'What went wrong' is a run problem (those mean Re-run, not Reject).
- I recorded my decision (see below).

Decision recorded: none yet. To record one, run make decide

REPORT=poc-main/reports/daily/2026-11-12/pytest-15380 DECISION=approve (or reject,

approve_with_notes, re_run).

Rerun command

make check PR=https://github.com/pytest-dev/pytest/pull/15380

Evidence links

- Pull request pytest-dev/pytest#15380
- record.json
- pr.diff
- setup-main.log
- setup-pr.log
- git.log
- pytest-main-1.log
- pytest-pr-1.log
- pytest-main-2.log
- pytest-pr-2.log

• pytest-main-3.log

• pytest-pr-3.log

• probe-model-probe1-main-1.jsonl

• probe-model-probe1-pr-1.jsonl

Glossary

| Term | Meaning

| main | The library as it is today, without the PR.

| PR | Pull request: a proposed change to the library.

| run | One full pass of the selected tests. We do 3 per side, alternating main and PR.

| flaky | A test that gives different results when nothing changed; it is ignored as evidence.

| log line | A line in a pytest-*.log file: the raw output the verdict is based on.

| route | How the two sides were built: nightly+patch, source-build or source-pair (see docs/OPERATIONS.md).

| changed lines covered | How many of the PR's changed, executable Python lines ran during the PR side's tests.

| re-parametrized | A test whose parameter ids changed (renamed cases); the new ids are judged on their own.

rebuilt from cleaned text

Written 2026-11-12 12:34 IST